

5.3 THE QUANTITY THEORY OF MONEY (QTM)

Multiple Choice Questions

1990/AL/II/14

Consider the following data of a hypothetical economy:

Year	Money Stock (\$ million)	GDP (\$ million, current prices)
1	45	400
2	50	600

Which of the following statements about the economy must be correct?

- A. There was a fall in real GDP.
- B. There was a fall in the general price level.
- C. There was a rise in the income velocity of money.
- D. There were fluctuations in the demand for money.

1990/AL/II/26

The Quantity Theory implies that in the short-run a change in the money stock will affect

- A. real output only.
- B. the price level only.
- C. both the real output and the price level.
- D. the income velocity of money.

1991/AL/II/14

During Christmas and Chinese New Year, people tend to carry more cash. Assuming banks do not hold excess reserves, there will be

- A. a decrease in the money supply.
- B. an increase in the money supply.
- C. no change in the money supply but a higher velocity of circulation.
- D. no change in the money supply but a higher demand for goods and services.

1991/AL/II/16

Which of the following ratios gives the correct definition of the velocity of the circulation of money?

- A. nominal GNP to the stock of money
- B. nominal GNP to the flow of money
- C. real GNP to the stock of money
- D. real GNP to the flow of money

1992/AL/II/20

The equation of exchange

- (1) is an identity which cannot be refuted.
 - (2) is valid only when the economy is at full employment.
 - (3) predicts that money affects both the price level and real output in the short run and only real output in the long run.
 - (4) state that total expenditure is always equal to total output value.
- A. (1) and (3) only
 - B. (1) and (4) only
 - C. (2) and (3) only
 - D. (2) and (4) only

1994/AL/II/17

According to the equation of exchange, if the money supply is fixed at a time when both the nominal and the real income are increasing, we can conclude that

- A. full employment is achieved.
- B. the unemployment rate is rising.
- C. the velocity of the circulation of money is falling.
- D. the velocity of the circulation of money is rising.

1994/AL/II/22

Given that the velocity of money and the real output remain unchanged, an increase in the money supply

- A. will decrease the nominal rate of interest.
- B. decrease the inflation rate.
- C. increase the real rate of interest.
- D. increase the price level.

1994/AL/II/24

Inflation will occur when

- A. the government carries out expansionary policies.
- B. the growth rate of money is greater than the growth rate of real output.
- C. the income of the economy grows.
- D. the prices of some of the goods start to rise.

1995/AL/II/18

The Quantity Theory of Money is

- A. always true because total expenditure are always equal to total output.
- B. valid because the price level is rigid in the short run.
- C. valid because it is an identity.
- D. None of the above

1996/AL/II/14

Which of the following statements about the quantity theory of money are correct?

- (1) Nominal income increases at the same rate as money supply.
 - (2) Velocity of circulation of money is assumed to be constant.
 - (3) The government cannot change the money supply.
- A. (1) and (2) only.
 - B. (1) and (3) only.
 - C. (2) and (3) only.
 - D. All of the above.

1996/AL/II/20

If the velocity of circulation of money is constant, an increase in the money supply will lead to an increase in

- A. the price level.
- B. real output.
- C. nominal value of output.
- D. None of the above

1997/AL/II/12

The classical quantity theory of money

- (1) predicts that a change in the money supply affects the price level only.
 - (2) is always true.
 - (3) assumes that the velocity of money circulation is constant.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3) only

1997/AL/II/22

The equation of exchange

- A. assumes that the velocity of money circulation is predictable.
- B. regards real output as more or less constant.
- C. explains the effects of changes in the stock of money on the price level.
- D. cannot be refuted.

1998/AL/II/14

According to the equation of exchange, the money supply grows more slowly than the nominal gross national product (GNP), we can conclude that

- A. the inflation rate is rising.
- B. the unemployment rate is rising.
- C. the velocity of circulation is rising.
- D. the growth rate of real GNP is higher than the inflation rate.

1999/AL/II/14

The equation of exchange says that

- A. the velocity of circulation of money is constant.
- B. nominal income will increase at the same rate as money supply.
- C. the economy is producing at the full employment level.
- D. None of the above

2000/AL/II/16

According to the equation of exchange, if money supply is fixed, which of the following is correct?

- A. Price level must be constant.
- B. Both price level and real output must be constant.
- C. The velocity of circulation of money will change at the same rate as real output.
- D. None of the above

2000/AL/II/22

Which of the following statements concerning money supply and inflation is correct?

- A. Any upward pressure on price level must be the result of an increase in money supply.
- B. An increase in aggregate demand must be accompanied by an increase in money supply and inflation rate.
- C. There can never be a simultaneous decrease in real output and increase in price level if the money supply does not decrease.
- D. None of the above

2001/AL/II/23

According to the equation of exchange, when the money supply is fixed,

- A. the growth rate of nominal income is constant.
- B. the growth rate of real output is the same as the inflation rate.
- C. the growth rate of nominal income is the same as the growth rate of the velocity of circulation of money.
- D. the inflation rate is zero.

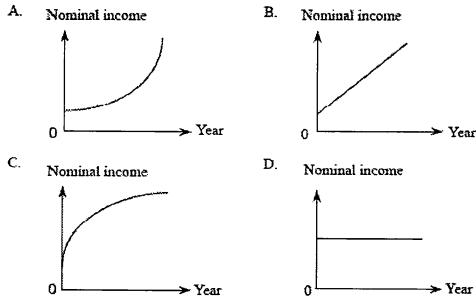
2002/AL/II/19

According to the equation of exchange, if money supply grows at the same rate as nominal income, we can conclude that

- A. the price level is rising.
- B. real output is constant.
- C. the velocity of circulation of money is constant.
- D. All of the above

2002/AL/II/23

If the velocity of circulation of money is constant and the money supply increases by a fixed amount each year, which of the following diagrams shows the value of nominal income over time?



2003/AL/II/11

According to the equation of exchange, if money supply grows during a time of deflation, it is possible that

- A. the velocity of circulation of money and real output decline at the same rate.
- B. the velocity of circulation of money and real output grow at the same rate.
- C. the growth rate of the velocity of circulation of money is higher than that of real output.
- D. the growth rate of the velocity of circulation of money is less than that of real output.

2004/AL/II/19

Consider the following data on an economy:

Year	Money Supply (\$ million)	GDP (\$ million, current prices)
1	45	400
2	50	600

From year 1 to year 2, which of the following statements about the economy must be correct?

- A. There was a rise in real GDP.
- B. There was a rise in the general price level.
- C. There was a rise in the velocity of circulation of money.
- D. The demand for money curve has shifted to the right.

2005/AL/II/14

The ratio of the value of transactions in an economy during a given period to the stock of money in that period measures

- A. money demand.
- B. the rate of growth of money supply.
- C. the velocity of circulation of money.
- D. the money multiplier.

2005/AL/II/18

Suppose the velocity of circulation of money is held constant. According to the equation of exchange, if the growth rate of money supply is smaller than that of the real gross national product (GNP), we can conclude that

- A. there is deflation.
- B. the inflation rate is falling.
- C. the unemployment rate is rising.
- D. the growth rate of real GNP is higher than the inflation rate.

2005/AL/II/21

When the velocity of circulation of money is constant, an increase in the money supply will lead to

- A. inflation.
- B. deflation.
- C. an increase in real output.
- D. an increase in nominal output.

2007/AL/II/11

An increase in the velocity of circulation of money will

- A. have no effect on real output whether or not money supply and price level are constant.
- B. lead to a rise in nominal output in the absence of changes in money supply.
- C. result in an increase in price level if the economy attains full employment.
- D. result in decrease in money supply if price level grows at the same rate as real output.

2008/AL/II/18

Which of the following statements about the velocity of circulation of money is correct?

- A. Velocity of circulation of money is measured by the ratio of nominal gross domestic product (GDP) to nominal money stock.
- B. The growth rate of velocity of money can be positive in some years and negative in others.
- C. The growth rate of velocity of circulation of money is assumed to be zero in the classical quantity theory of money.
- D. All of the above

2009/AL/II/22

Consider the following data of an economy.

	Year 1	Year 2
Money supply (\$)	40 million	48 million
Velocity of circulation of money	5	5.5
Real output (unit of goods)	20 million	24 million

According to the equation of exchange, what is the inflation rate of Year 2?

- A. 8%
- B. 9.1%
- C. 10%
- D. 11%

2011/AL/II/24

If the real output is decreasing at a faster rate than the velocity of circulation of money,

- A. money supply will be increasing at a rate slower than that of the price level.
- B. money supply will be decreasing at a rate slower than that of the price level.
- C. money supply will increase when there is inflation.
- D. there will be deflation when money supply remains unchanged.

2012/AL/II/16

If an economy attains full employment, an increase in money supply will increase _____.

- A. the nominal interest rate
- B. the real interest rate
- C. the level of nominal output.
- D. the level of real output

2012/AL/II/20

Which of the following statements about the equation of exchange is correct?

- A. It predicts that money is neutral.
- B. It still holds even if there is a change in institutional factors.
- C. It predicts that a 1% increase in money supply leads to a 1% increase in the price level.
- D. It assumes that the technological level is constant.

2013/DSE/I/32

The real output of an economy grows at a lower rate than the money supply. According to the quantity theory of money, we can conclude that

- (1) the price level is increasing.
 - (2) the unemployment rate is decreasing.
 - (3) the velocity of circulation of money is decreasing.
 - (4) the nominal output is increasing.
- A. (1) and (2) only
 - B. (1) and (4) only
 - C. (2) and (3) only
 - D. (3) and (4) only

2014/DSE/I/37

Suppose the velocity of circulation of money remains constant. If a central bank reduces money supply by 10%,

- A. the price level will decrease by 10%.
- B. real output will decrease by 10%.
- C. nominal output may decrease by less than 10%.
- D. real output may decrease by less than 10%.

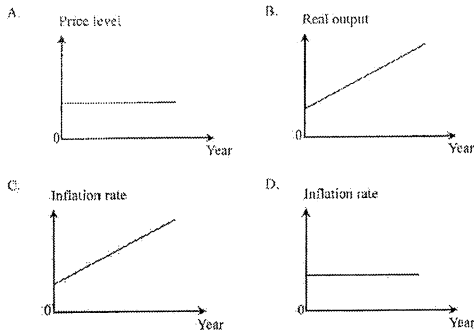
2015/DSE/I/33

The quantity theory of money implies that

- A. an increase in money supply will lead to a rise in price level when real output is falling.
- B. the economy is always at the full employment output level.
- C. an increase in money supply will lead to a decrease in the velocity of circulation of money when real output is kept constant.
- D. an increase in money supply will lead to a same-proportional increase in real output.

2016/DSE/I/34

According to the classical quantity theory of money, which of the following diagrams is correct when money supply is growing at a constant percentage each year?



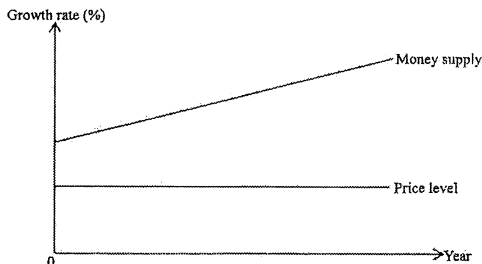
2017/DSE/I/36

According to the quantity theory of money, if a central bank increases money supply,

- A. the price level will increase.
- B. real output will remain constant.
- C. nominal output will increase.
- D. the unemployment rate will decrease.

2018/DSE/I/37

Refer to the following diagram.



According to the quantity theory of money, which of the following statements is correct?

- A. The inflation rate is zero.
- B. The growth rate of real output is increasing.
- C. The growth rate of money supply is constant.
- D. The growth rate of nominal income is constant.

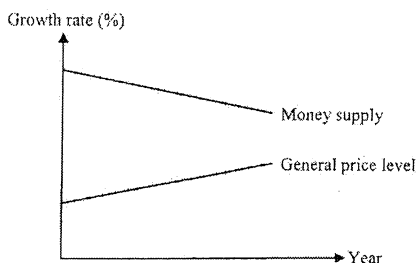
2019/DSE/I/35

Which of the following statements about quantity theory of money are correct?

- (1) If money supply is growing too fast relative to the supply of goods in the economy, inflation will occur.
 - (2) If money supply increases, nominal output will increase by the same percentage.
 - (3) If inflation rate is constant, real output will grow at the same percentage as money supply.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2021/DSE/I/38

Study the following diagram.



According to the quantity theory of money, which of the following statements is correct?

- A. Real output is increasing.
- B. Nominal output is decreasing.
- C. Money supply is decreasing.
- D. The velocity of circulation of money is decreasing.

2023/DSE/I/37

According to the classical quantity theory of money, if the money supply of an economy increases by 8%, _____.

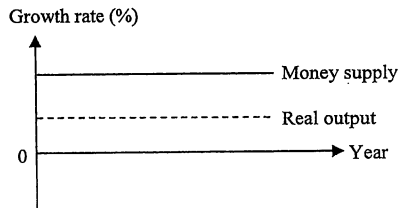
- A. the government's income tax revenue will increase by 8%
- B. the nominal output will increase by 8%
- C. the real output will increase by 8%
- D. the price level may increase by less than 8%

2024/DSE/I/37

Which of the following about the quantity theory of money is correct?

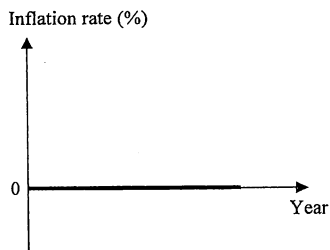
- A. An increase in money supply will not affect the nominal output in the long run.
- B. If the money supply growth rate is constant, the price level will also be constant in the long run.
- C. If the price level is constant, a decrease in real output will lead to a decrease in money supply.
- D. If the real output growth rate is zero, the price level growth rate will be equal to the money supply growth rate.

Study the following diagram.

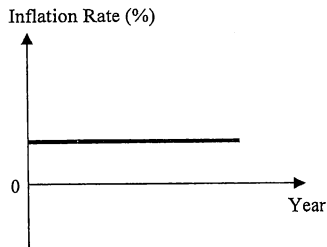


Based on the above information, which of the following is an implication of the quantity theory of money?

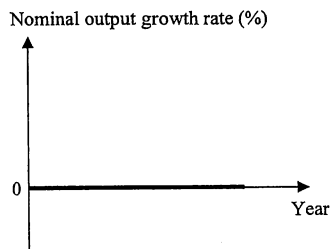
A.



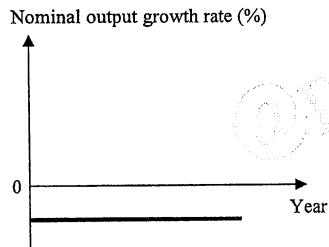
B.



C.



D.



Short & Structured Questions

1990/AL/II/8(a)(ii)

Explain whether inflation would occur in the following situations:

a continuous increase in the money supply when the economy is also growing.

(8 marks)

1991/AL/II/2

'The classical Quantity Theory of Money is a theory of full employment.' Do you agree? Explain your answer. (10 marks)

1993/AL/II/4

If the national income and the money supply of a country grow at 5% and 10% a year respectively, what is the rate of

inflation predicted by the Quantity Theory of Money? Explain.

(8 marks)

1999/AL/II/4

(a) Milton Friedman once made the statement, 'Inflation is always and everywhere a *monetary* phenomenon.' In what sense do you think inflation is related to monetary factors? (4 marks)

(b) Using the quantity theory of money, explain why, in the absence of changes in money supply, inflation can also be 'caused' by real factors such as changes in real output. (4 marks)

2000/AL/II/8

(a) 'If the velocity of circulation of money is constant, we can predict the impact of changes in money supply on nominal income. However, to know the impact of changes in money supply on price level, we need an additional assumption on the behaviour of real income.' Discuss. (6 marks)

(b) Consider the following table.

Year	1	2	3
Money Supply	\$100	\$110	\$121

Suppose the velocity of circulation of money is constant.

(i) Find the growth rate of nominal income in year 2 and year 3. (3 marks)

(ii) Suppose the growth rates of real income in year 2 and year 3 are 0% and 3% respectively. Find the inflation rates in year 2 and year 3. (2 marks)

(c) Based on the above answers, explain the rationale behind the policy of setting the growth of money supply at a specific level. (2 marks)

2002/AL/II/2(a)

State whether each of the following statements is true or false, and provide a brief explanation to support your answer.

'The quantity theory of money predicts that inflation is not necessarily a consequence of money growth.' (5 marks)

2007/AL/II/3(b)

What is meant by the velocity of circulation of money? How can we measure such velocity in practice?

(2 marks)

2010/AL/II/2(c)

Explain why the product of the nominal stock of money (M) and the velocity of circulation (V) is always equal to the product of the price level (P) and real income (y) – i.e., $MV = Py$. How can we use this relation as a theory (to identify the various sources) of inflation? (6 marks)

2012/DSE/II/13(d)

The central bank injects \$200 million newly-printed banknotes into the banking system.

Based on the quantity theory of money, predict the effect of the monetary policy on the price level. State the assumption(s) needed to reach your conclusion. (4 marks)

2018/DSE/II/7(b)

Based on the quantity theory of money, predict the effect of an open market purchase of government bonds by the central bank on the price level. State the assumption(s) needed to reach your conclusion. (4 marks)

2020/DSE/II/4C

Study the following information of an economy.

Year	Consumer price index (CPI)
2018	110
2019	117

- (c) 'The continuous open market purchase of government bonds by the central bank *is* the cause of inflation in the economy.' Justify the statement with reference to the classical quantity theory of money. (4 marks)

2022/DSE/II/6a

- (a) 'Deflation will occur if the amount of goods and services produced is growing faster than money supply,' Explain this statement in terms of the quantity theory of money. (4 marks)

2023/DSE/II/12b

Source C: Money supply in the US

	May 2020	May 2021
Money supply (USD Billion)	17766.9	20 319.6

- (b) Refer to Source C. According to the quantity theory of money, explain under what condition the change in money supply as shown in the source would result in an increase in the general price level in the short run. (3 marks)